

Beauty & Wellness Data Science Orientation

1. Introduction

The Beauty & Wellness industry is one of the fastest growing sectors today. Skincare, haircare, spa services, and organic products are seeing increasing demand every year. Customers are becoming more conscious about their health and appearance. With this growth comes a challenge: brands need to understand which products customers prefer and when demand will rise. Data science can play a powerful role here by helping businesses analyze customer behavior and predict future trends.

2. Problem Statement

The main problem addressed in this project is: *"Predicting customer purchase trends in skincare products."* Brands often struggle to identify which products will sell more and when demand will peak. If these trends can be predicted, inventory management and marketing strategies can be planned more effectively.

3. Objectives

- Analyze customer purchase behavior.
- Predict future demand trends for skincare products.
- Provide actionable insights to improve inventory planning and marketing strategies.

4. Research Questions & Hypotheses

- **RQ1:** Does age and gender influence skincare product preferences?
- **RQ2:** Do seasonal factors (summer/winter) impact product demand?
- **Hypothesis:** Younger customers prefer organic skincare products more than older customers.

5. Variables & Data Sources

- **Customer Demographics:** Age, Gender, Location
- **Purchase History:** Product category, frequency, spending amount
- **External Factors:** Season, festival periods, social media trends
- **Data Sources:** E-commerce sales data, customer surveys, social media reviews

6. Analysis Plan

- **Exploratory Data Analysis (EDA):** Identify patterns and anomalies.
- **Predictive Modeling:** Use regression or classification models.
- **Recommendation System:** Apply collaborative filtering techniques.
- **Sentiment Analysis:** Study social media reviews to understand brand perception.

7. Expected Outcomes

- Prediction of top-selling skincare products.
- Insights into customer segmentation (e.g., age group preferences).
- Clear recommendations for marketing and inventory planning.

8. Conclusion

Applying data science in the Beauty & Wellness industry provides a strong advantage. This project will create a framework to help brands understand customer behavior and predict demand more accurately. The orientation task serves as a solid foundation for future detailed analysis and practical implementation.